observable averages diverge and we do not recover correct sampling in the limit of zero stepsize.

The symmetric schemes considered in the previous section are consistent however, each with their own limiting schemes corresponding to taking the limit of infinite friction in their update equations. As the momenta are continually redrawn from a normal distribution, we focus on the evolution of the configurations themselves. For the method denoted ⟦OBABO⟧, we have the limiting scheme

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &= \sqrt{k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{1/2}\mathbf{R}_n,\\
\hat{\boldsymbol{p}}_{n+1/2} &= \boldsymbol{p}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_n),\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\hat{\boldsymbol{p}}_{n+1/2},\\
\hat{\boldsymbol{p}}_{n+1} &= \hat{\boldsymbol{p}}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}),\\
\boldsymbol{p}_{n+1} &= \sqrt{k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{1/2}\mathbf{R}_{n+1/2},
\end{aligned}$$

where $\mathbf{R}_n$ and $\mathbf{R}_{n+1/2}$ are independent normal random vectors with mean zero and unit variance. It is clear that if our only desire is to make use of the position data $\boldsymbol{q}$, then the final two steps are redundant in this algorithm, as we update the momentum vector only to redraw it at the beginning of the next iteration. Any information about the initial momentum vector $\boldsymbol{p}_n$ is unused in the scheme, making it simple to rewrite solely in terms of the position vector, giving us

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n - \frac{h^2}{2}\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}_n) + h\sqrt{k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{-1/2}\mathbf{R}_n. \tag{7.38}$$

Denoting $\tau = h^2/2$, the update scheme (7.38) becomes the Euler-Maruyama scheme

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n - \tau\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}_n) + \sqrt{2\tau k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{-1/2}\mathbf{R}_n, \tag{7.39}$$

which is known to give an error in long-time averages $\mathcal{O}(\tau) = \mathcal{O}(h^2)$, just as our analysis suggests.

If we instead consider the ⟦BAOAB⟧ scheme we find that unlike the ⟦OBABO⟧ scheme we no longer have redundant steps:

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &= \boldsymbol{p}_n - (h/2)\nabla U(\boldsymbol{q}_n),\\
\boldsymbol{q}_{n+1/2} &= \boldsymbol{q}_n + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2},\\
\hat{\boldsymbol{p}}_{n+1/2} &= \sqrt{k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{1/2}\mathbf{R}_n,\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_{n+1/2} + (h/2)\boldsymbol{M}^{-1}\hat{\boldsymbol{p}}_{n+1/2},\\
\boldsymbol{p}_{n+1} &= \hat{\boldsymbol{p}}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}),
\end{aligned}$$

where $\mathbf{R}_n$is an independent normal random vector with mean zero and unit variance. Expressing this as an update solely for the configurations, we obtain

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n - \frac{h^2}{2}\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}_n) + \frac{h}{2}\sqrt{k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{-1/2}\left(\mathbf{R}_n + \mathbf{R}_{n-1}\right),$$

and substituting $\tau = h^2/2$ gives the BAOAB *Limit Method* (Leimkuhler-Matthews method) [221]

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n - \tau\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}_n) + \sqrt{2\tau k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{-1/2}\left(\frac{\mathbf{R}_n + \mathbf{R}_{n-1}}{2}\right). \tag{7.40}$$

As we are in the overdamped regime, we would expect that this scheme should have an $\mathcal{O}(h^4) = \mathcal{O}(\tau^2)$ error in configurational averages, while inheriting the other properties of the ⟦BAOAB⟧ scheme for the harmonic and perturbed harmonic models considered in Sects. 7.4 and 7.5. Moreover, since we completely remove the second order term in the Langevin dynamics configurational density expansion, we expect to observe this behavior across all values of h and τ. This is quite a surprising result, given the qualitative similarity between Eqs. (7.39) and (7.40).

The benefit of (7.40) over the Euler-Maruyama scheme is that we can get second order accuracy in averages with only a single evaluation of the force at each timestep. However, our analysis presented here is only for the infinite time case. For averages computed in finite time, work in [224] has shown that (7.40) does indeed give a first order error, albeit one that vanishes exponentially fast in time as the simulation progresses. Hence for suitably long simulations, the observed error will be $\mathcal{O}(h^2)$.

We can think of the noise process in (7.40) as corresponding to a colored noise which has characteristics that directly depend on the stepsize, although the noise decorrelates in just a couple of timesteps. This is therefore no longer a Markov process, however it can be reformulated as such if one considers the appropriate extended space (or keeping the momenta $\boldsymbol{p}_n$).

7.11 Applying the Schemes to a Biomolecule

Numerical analysis of a scheme should always be supported (and informed by) thorough numerical testing. Often salient features of a discretization method can be missed because of a shallow scope of experimentation. Our goal will be to evaluate the efficacy of the numerical schemes for Langevin dynamics presented thus far. We shall look at the invariant (long-time) averages resulting from computations using the four symmetric Langevin integrators that we derived using our splitting, for which results for the harmonic oscillator are detailed in (7.15), and for which a more general error analysis was considered in Sect. 7.9.2. Additionally we will compare their results to the performance of the standard SPV and BBK schemes considered in Sect. 7.3.2.